

11-Frontend-Architecture.md

Healthcare Clinical Intelligence Platform

Enterprise Frontend Architecture

Version: 1.0

Status: Approved Design

Mission

Build a healthcare-grade SaaS platform that provides:

- Exceptional User Experience
- Enterprise Administration
- AI Transparency
- Knowledge Governance
- Full Observability
- Multi-Tenant Operations

The frontend must feel closer to:

- Glean
- OpenEvidence
- Perplexity Enterprise
- Harvey AI
- Cursor
- Linear

Not a simple chatbot.

Frontend Technology Stack

Framework

- Next.js

Language

- TypeScript

UI

- ShadCN UI

Styling

- TailwindCSS

State Management

- Zustand

Server State

- TanStack Query

Forms

- React Hook Form

Validation

- Zod

Charts

- Recharts

Tables

- TanStack Table

Authentication

- Supabase Auth

Application Structure

```
Platform
|
├─ Authentication
├─ Workspace
├─ Knowledge Base
├─ AI Assistant
├─ Documents
├─ Evaluation
```

- └─ Observability
- └─ Administration
- └─ Settings

User Types

Platform Super Admin

Organization Admin

Department Admin

Doctor

Pharmacist

Researcher

Viewer

Main Navigation

Dashboard

AI Assistant

Knowledge Bases

Documents

Memory

Evaluations

Observability

Administration

Settings

Authentication Module

Features

- Login
 - Signup
 - SSO Ready
 - MFA Ready
 - Password Reset
 - Session Management
-

Dashboard

Purpose

Single overview of platform health.

Widgets

Recent Queries

Document Status

Approval Queue

System Health

Evaluation Metrics

Organization Activity

Cost Summary

Recent Alerts

AI Assistant Module

Most Important Screen

Layout

```
-----  
Sidebar  
|  
Chat Area  
|  
Evidence Panel  
|  
Trace Panel  
-----
```

AI Assistant Features

Chat Interface

Conversation History

Source Citations

Confidence Scores

Suggested Follow-Ups

Export Responses

Copy Responses

Share Responses

Evidence Panel

Displays

Retrieved Chunks

Source Documents

Source Diversity

Evidence Strength

Confidence Scores

Citation Viewer

Displays

Document

Page

Section

Version

Publication Date

Agent Trace Viewer

Displays

Planner Decisions

Retrieval Strategy

GraphRAG Usage

Validation Results

Safety Decisions

Final Confidence

Knowledge Base Module

Purpose

Manage organizational knowledge.

Knowledge Base Features

Create KB

Edit KB

Archive KB

Department Assignment

Role Assignment

Usage Metrics

Knowledge Base Types

Organization

Department

Personal

External

Knowledge Base Analytics

Documents

Queries

Users

Retrieval Performance

Cost

Document Management Module

Purpose

Govern document lifecycle.

Document Features

Upload

Review

Approve

Reject

Archive

Version History

Reindex

Document Detail Page

Displays

Metadata

Chunks

Entities

Versions

Approvals

Usage Statistics

Document Approval Workflow

Draft
↓
Pending Review
↓
Approved
↓
Archived

Version Management

Displays

V1

V2

V3

Differences

Approval History

Memory Module

Purpose

Memory governance.

User Memory View

Preferences

Interaction Patterns

Saved Memories

Department Memory View

Department Knowledge

Department Policies

Organization Memory View

Institutional Knowledge

Approved Memories

Agent Memory Review

Displays

Memory Candidates

Approval Queue

Rejections

Evaluation Module

Purpose

Monitor quality.

Evaluation Dashboard

Displays

Context Precision

Context Recall

Faithfulness

Citation Coverage

Hallucination Rate

Safety Accuracy

Golden Dataset Module

Displays

Benchmarks

Regression Results

Model Comparisons

Feedback Dashboard

Displays

Helpful

Not Helpful

Hallucinations

Wrong Citations

Incomplete Answers

Observability Module

Purpose

Platform transparency.

AI Observability Dashboard

Displays

Agent Activity

Tool Calls

Retrieval Metrics

Model Usage

Retrieval Dashboard

Displays

Query Volume

Source Diversity

GraphRAG Usage

Retrieval Confidence

Retry Rate

Cost Dashboard

Displays

Cost Per Query

Cost Per User

Cost Per Organization

Cost Per Model

Latency Dashboard

Displays

API Latency

Retrieval Latency

Agent Latency

LLM Latency

Administration Module

Purpose

Organization management.

Organization Management

Organizations

Departments

Users

Roles

Permissions

User Management

Invite Users

Deactivate Users

Role Assignment

Department Assignment

API Key Management

Create Keys

Rotate Keys

Disable Keys

Usage Monitoring

Audit Dashboard

Displays

Logins

Approvals

Retrieval Events

Security Events

Admin Actions

Security Dashboard

Displays

RBAC Violations

Access Denials

Failed Logins

API Abuse

Search Experience

Global Search

Searches:

Documents

Knowledge Bases

Conversations

Users

Evaluations

Notification Center

Displays

Approval Requests

Failed Ingestion Jobs

Security Alerts

System Notifications

Settings Module

Organization Settings

Department Settings

AI Settings

Security Settings

Billing Settings

Retention Policies

Multi-Tenant UX

Every page filtered by:

Organization

Department

Role

Knowledge Base

Accessibility

WCAG Compliance

Keyboard Navigation

Screen Reader Support

High Contrast Mode

Mobile Strategy

Primary:

Desktop First

Secondary:

Tablet

Limited Mobile Support

Frontend Observability

Track

Page Views

User Actions

Errors

Performance

Feature Usage

Design Principles

Transparency

Explainability

Minimal Cognitive Load

Fast Navigation

Trust Through Evidence

Governance Visibility

Success Metrics

Page Load < 2s

Time To First Response < 5s

User Satisfaction > 4.5/5

Feature Adoption > 70%

Admin Task Completion > 95%

Future Features

Voice Interface

Collaborative Research

Clinical Workspaces

Shared Investigations

Real-Time Co-Pilot

Mobile Application

Final Principle

The UI should make complex AI systems feel simple.

Users should always understand:

What the system knows.

Why it answered.

Where the evidence came from.

How confident it is.